Using Cosmological Simulations to Investigate the Suppression of Dark Matter Halo Growth by Massive Neutrinos

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/s/ Geoffrey Zheng

Abstract

Neutrinos are the lightest standard-model fermions, and neutrino oscillation experiments have now measured the differences of their masses squared to high precision. However, measuring absolute neutrino masses in the laboratory is challenging due to the large difference in mass scales between neutrinos and other standard model particles. Fortunately, massive neutrinos also affect the growth of large-scale structure in our universe. Therefore, by studying how neutrinos interact with matter in large cosmological simulations and comparing the results with observations, we can obtain bounds on the absolute mass of the neutrino. In this junior paper, we study two fast neutrino simulation algorithms used to simulate the growth of dark matter halos in the presence of massive neutrinos: the Linear Response Approximation (LRA) algorithm and the Hybrid Method. We first assess the validity of the Hybrid method in simulating dark matter halo growth by cross-checking the simulation results against the results obtained from using the already-tested LRA algorithm and the N-body particle-based "truth" method. We then use all three simulation methods to investigate the effects of increasing neutrino mass on halo growth. The total neutrino mass in each simulation run is altered and the halo mass function is computed from the corresponding simulation data. Thus we determine the halo mass range where suppression in growth due to neutrinos occurs. The results of this paper help determine whether the Hybrid model can be used as a new standard tool to model massive neutrinos and also provides more information about halo mass suppression due to neutrinos, which may allow cosmologists to finally constrain the value of the neutrino mass.

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1 Introduction

The Standard Model of particle physics is the current accepted theory about three of the four fundamental forces of Nature: electromagnetic force, weak force, and strong force. Broadly speaking, the electromagnetic force is responsible for almost all force interactions we are familiar with in daily life, from frictional forces to the usual electrical forces in circuits. The weak force is responsible for the radioactive decay of atoms, and the strong force is responsible for the binding of the nuclei of atoms. In addition, the Standard Model categorizes all known elementary particles, which are particles with no sub-structure (i.e. not made up of any other particles) [1]. The Standard Model has not yet successfully incorporated the gravitational force.

According to the Standard Model, elementary fermions are the building blocks of baryonic matter, meaning they obey the Fermi-Dirac statistics in describing their number distribution over allowed energy levels:

$$\bar{n}(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1}, \tag{1}$$

where E is the energy, μ is the chemical potential, k_B is the Boltzmann constant, and T is the temperature. Within this classification, fermions are split into two classes: quarks and leptons. There are 6 quarks, which cannot exist on their own; they combine to form more massive particles classified as mesons and baryons. In particular, protons and neutrons are the baryons that combine to form atoms. There are also 6 leptons, which do not undergo strong force interactions and form the lightest class of matter particle that can exist on its own. In this paper we focus on a particular type of lepton: the neutrino.

1.1 Neutrinos in Particle Physics

The neutrino is a spin-1/2 fermion that is electrically neutral and has extremely small mass. Neutrinos come in 3 flavors, which make up 3 of the 6 leptons in the Standard Model: electron neutrino ν_{e}, muon neutrino ν_μ, and tau neutrino ν_τ. More formally, we can say that $|\nu_{\mathrm{e}}\rangle$, $|\nu_\mu\rangle$, and $|\nu_\tau\rangle$ make up the three flavor eigenstates of the neutrino. Each neutrino has an associated antimatter particle called an antineutrino. Because the neutrino is electrically neutral, it only interacts via the weak force and gravity [1].

The neutrino, while a part of the Standard Model, has long been the source of various puzzles and discrepancies that have led to the development of new physics. As early as the 1960s, the astrophysicist John Bahcall developed the theory of the Standard Solar Model, which among other things, predicted the flux of solar neutrinos on Earth. However, the theoretical prediction of the flux was approximately 3 times more than

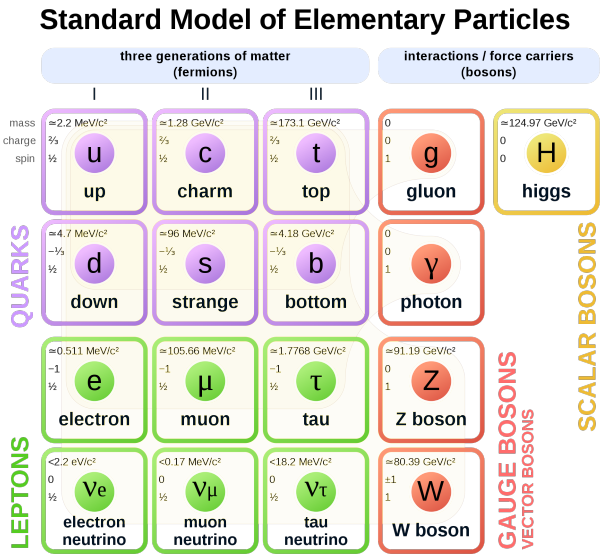

Figure 1: Schematic diagram of the Standard Model of Elementary Particles. Neutrinos span 3 generations of matter and are classified as leptons, which themselves are classified as fermions. Image taken from Wikimedia Commons (Public Domain): https://commons.wikimedia.org/w/index.php?curid=4286964.

what experiments such as the Homestake experiment detected on Earth. The puzzle was solved when physicists realized that the mechanism of neutrino oscillations could help explain the flux deficit. Theoretically, neutrino oscillations occur if the flavor eigenstates of the neutrino do not correspond to the mass eigenstates of the neutrino, which we denote $|m_1\rangle$, $|m_2\rangle$, and $|m_3\rangle$. That is, all three neutrino masses consist of different linear combinations of the three neutrino flavors. In such a scenario, quantum mechanics dictates that the phases of the mass eigenstates would change at different rates in a periodic fashion, meaning that as the neutrino propagates through space, it should oscillate between different flavors. The first observation of neutrino oscillations by a terrestrial (Earth-based) neutrino experiment occurred at the Super-Kamiokande Observatory in Japan [2] in 1998. Experimental detection of neutrino oscillations thus contributed to new physics in two ways. First, it vindicated the Solar Model of John Bahcall, because the original detectors used to detect neutrinos could only detect one flavor, thus missing 2/3 of the flux that comes from neutrinos oscillating into the other flavors. Second, the discovery of neutrino oscillations directly implied that neutrinos have nonzero mass.

Importantly, although the experiments mentioned above have shown that neutrinos have mass, we can actually only measure the differences of the squares of the three masses

in the oscillation experiments. The most recent values are given by [3]:

$$\Delta m_{12}^2 \equiv m_2^2 - m_1^2 = 7.37 \times 10^{-5} \text{eV}^2. \tag{2}$$

$$|\Delta m_{13}^2| \equiv |m_3^2 - (m_1^2 + m_2^2)/2| = 2.525 \times 10^{-3} \text{eV}^2. \tag{3}$$

We do not currently know the individual masses of each mass eigenstate. Further, because we only know the absolute value of $|\Delta m_{13}^2|$ and not the actual sign, two possible neutrino mass hierarchies arise. The first hierarchy is known as the normal hierarchy, and assumes that $m_1 \ll m_2 < m_3$. The second hierarchy is known as the inverted hierarchy, and assumes $m_3 \ll m_1 < m_2$. According to the normal hierarchy, it is predicted that the total sum of neutrino mass $M \equiv m_1 + m_2 + m_3$ is approximately $M \gtrsim 0.06$ eV, while according to the inverted hierarchy, it is predicted that $M \gtrsim 0.1$ eV [4].

It is of key interest to us to be able to obtain stricter constraints on the absolute mass of the neutrino, so that we can work towards furthering our understanding of neutrino physics. Furthermore, according to the current Standard Model, the neutrino is massless, and this cannot be definitively changed until we obtain a definitive measurement of the absolute neutrino mass. However, whereas current terrestrial-based neutrino experiments cannot yet measure the absolute mass of the neutrino, current cosmological data can already provide constraints on neutrino properties that are not only complementary but competitive to terrestrial experiments [5].. For example, data from the Planck cosmology experiment shows that by analyzing cosmic microwave background (CMB) power spectra, lensing, and baryonic acoustic oscillations, we can obtain a constraint on the neutrino mass of $M \lesssim 0.12$ eV [6]. Therefore, to gain a further understanding of neutrino mass, we turn to cosmology.

1.2 Fundamentals of Cosmology

Cosmology is the study of the physical structure of our Universe at the largest scales. One of the central tenets of cosmology is that on the cosmological scale (distance scales of ~ 100 Mpc or greater), the universe is nearly perfectly homogeneous and isotropic [7]. Homogeneity means there is no preferred location in the universe, whereas isotropy means there is no preferred direction in the universe. This is known as the cosmological principle. At scales less than the cosmological scale, the universe is clearly inhomogeneous, as evidenced by the plethora of stars, galaxies, and other structures that pervade our universe.

The current accepted model of cosmology is known as the Lambda-Cold Dark Matter (LCDM) model. It states that the current content of the universe is approximately 5% baryonic matter, 25% non-baryonic dark matter, 70% dark energy, and trace

amounts of radiation. Baryonic matter refers to all matter composed of particles from the current Standard Model. Dark matter is non-baryonic and only interacts gravitationally with baryonic matter. Since we treat the dark matter as non-relativistic from matter-radiation equality onward, we call it cold dark matter (CDM). Dark energy is a gravitationally self-repulsive form of energy that is hypothesized to take the form of a cosmological constant, and is responsible for the accelerated expansion of the universe. Under the LCDM model, the universe is also assumed to be nearly perfectly flat. The most recent results from the Planck experiment confirm this, with $\Omega_m = 0.3111 \pm 0.0056$ and $\Omega_\Lambda = 0.6889 \pm 0.0056$ [6].

Because the universe is homogeneous and isotropic at large scales, the metric used in cosmology to describe spacetime is the Friedmann-Robertson-Walker (FRW) metric, which is given by (in Einstein notation):

$$ds^2 = -c^2 dt^2 + a^2(t) dx_i^2, \tag{4}$$

where dx_i^2 represents the metric of either flat, positively curved (spherical), or negatively curved (hyperbolic) space. $a(t)$ is the scale factor which parameterizes distances between objects in an expanding universe. By convention, a is normalized so that at the Big Bang singularity, $a = 0$ and at the present time, $a = 1$. Furthermore, a is related to the redshift z through the relation $a = \dfrac{1}{1+z}$, so that increasing redshift (going further back in time) corresponds to decreasing a.

At the cosmological scale, the physics is essentially governed by general relativity and gravity through the Einstein field equations:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{5}$$

where $R_{\mu\nu}$ is the Ricci curvature tensor, R is the scalar curvature, $g_{\mu\nu}$ is the metric tensor, and $T_{\mu\nu}$ is the stress-energy tensor. Substituting the FRW metric for $g_{\mu\nu}$ into the Einstein equations and taking the 00 (time) component gives the first Friedmann equation (setting $c = 1$):[1]

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \sum_i \rho_i - \frac{\kappa}{a^2}, \tag{6}$$

Here, H denotes the Hubble parameter, which parameterizes the expansion of the universe. G is Newton's gravitational constant. ρ_i denotes the different components of the matter-energy density, such as matter, radiation, and dark energy. κ parameterizes the curvature of the universe; positively curved space has $\kappa = +1$, negatively curved space has $\kappa = -1$, and flat space has $\kappa = 0$.

[1] For brevity and relevance, a full derivation of the Friedmann equations from the Einstein equations is not given here. A good, full derivation is found in [8], Ch. 8.

Using the cosmological principle and the FRW metric, we treat the matter-energy content of our universe as a perfect fluid. A perfect fluid is one that is completely characterized by its density and isotropic pressure. With this approximation and the conservation of energy in the form of the First Law of Thermodynamics, we obtain the Fluid Equation:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0, \tag{7}$$

where p denotes the pressure. Finally, we use the equation of state of a perfect fluid to relate its energy density to its pressure: $p = w\rho$, where w is a dimensionless parameter depending on the component in question. Together, the Friedmann equation and the Fluid Equation serve as the workhorse equations of cosmology and dictate the behavior of how different components of the universe evolved.

Using equation 7 and the equation of state, we can derive the evolution of the energy density of the various components. Plugging the equation of state into 7 yields:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + w\rho) = 0 \tag{8}$$

$$\frac{d\rho}{\rho} = -3(1+w)\frac{da}{a} \tag{9}$$

$$\rho(a) = \rho_0 a^{-3(1+w)}. \tag{10}$$

For matter domination, w = 0; for radiation domination, w = 1/3; for dark energy domination, w = −1. Hence, we see that $\rho \sim a^{-3}$ for matter, $\rho \sim a^{-4}$ for radiation, and $\rho \sim$ constant for dark energy. In other words, extrapolating back in time, the universe was first dominated by radiation, followed by matter, followed by dark energy in the present. This is shown qualitatively in Figure 2, where we use the fact that $z \sim 1/a$.

2 Massive Neutrinos in Cosmological Structure Formation

2.1 The Physics of Large Scale Structure Formation

An important phenomenon that cosmology seeks to explain is the formation of large-scale structure in our universe, such as galaxies, galaxy clusters, filaments, halos, and voids. Structure formation first arose from the amplification of primordial perturbations in the matter-energy density field of the early universe. The evolution of these perturbations due to the Hubble expansion and their own gravitational attraction can be derived by considering the usual perfect fluid approximation of the matter-energy content of our universe. For our fluid with density ρ, pressure p, and velocity $\vec{v}$, there are 3 governing equations:

$$\frac{\partial \rho}{\partial t} = -\nabla_r \cdot (\rho\vec{v}). \tag{11}$$

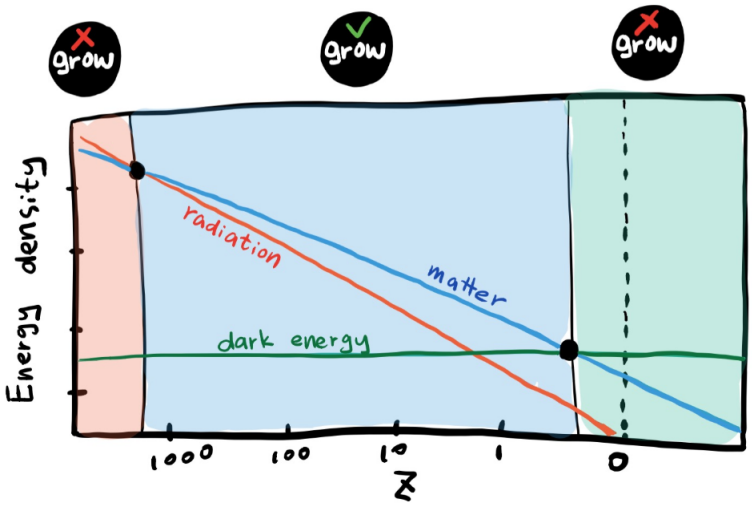

Figure 2: Evolution of the energy density as a function of redshift, with a logarithmic scale on the x-axis. Matter-radiation equality occurred at approximately $z = 3500$ and matter-dark energy equality occurred at approximately $z = 0.43$. Structure formation can only occur during the matter-dominated era. Diagram taken from [9].

$$\frac{d\vec{v}}{dt} + (\vec{v} \cdot \nabla_r)\vec{v} + \frac{\nabla_r p}{\rho} + \nabla_r \Phi = 0. \tag{12}$$

$$\nabla_r^2 \Phi = 4\pi G \rho. \tag{13}$$

Equation 11 is the continuity equation of a fluid; equation 12 is the Euler equation, which derives from the Navier-Stokes equation for a fluid with no viscosity; equation 13 is the Poisson equation for a gravitational source potential Φ. ∇_r denotes the gradient with respect to regular position coordinate $\vec{r}$. To decouple the scale factor from our position coordinates, we convert to co-moving position $\vec{x}$ through the relation $\vec{x} = a\vec{r}$, meaning the gradient is converted to the co-moving gradient as $\nabla_r = \frac{1}{a}\nabla_x$. We then linearize the perturbations in ρ, $\vec{v}$, and Φ to first order:

$$\rho = \rho_0 + \delta\rho; \qquad \vec{v} = H\vec{r} + \delta\vec{v}; \qquad \Phi = \Phi_0 + \delta\Phi. \tag{14}$$

Plugging these relations back into the governing fluid equations, converting to co-moving coordinates, and only keeping terms to first order yields the evolution of the density perturbations:[2]

$$\ddot{\delta} + 2H\dot{\delta} - \frac{c_s^2}{a^2}\nabla_x^2 \delta = 4\pi G \rho_0 \delta, \tag{15}$$

where $\delta \equiv \frac{\delta\rho}{\rho}$ is the relative growth of the perturbation and c_s is the speed of sound waves in the fluid. Note that equation 15 is valid so long as the perturbation δ is small; if $\delta \sim 1$, the structure has grown to sufficient size the linear approximation to

[2]See complete derivation in [10], Ch. 6 Section 3.

the perturbation is no longer valid, thus rendering the behavior to start to transition to non-linear.

We now consider the case of evolution of δ in the matter-dominated era. In the matter-dominated era, we assume that the entire energy density only contains a matter component, so $\rho \sim a^{-3}$. Thus by equation 6 (assuming a flat universe), we have:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0 = \frac{8\pi G}{3a^3}\rho_{m,0} \implies \dot{a} \sim a^{-1/2}. \tag{16}$$

Hence using separation of variables, $a \sim t^{2/3}$ in the matter-dominated era. Hence using the definition of H, we know $H \sim 2/(3t)$. Now, we guess an ansatz for δ of the form $\delta \sim t^{\alpha}$, plug the evolution of H into equation 15, and take the $\nabla_x^2\delta$ term to be negligible due to the non-relativistic nature of dark matter, to obtain:

$$\alpha(\alpha-1)t^{\alpha-2} + \frac{4}{3t}\alpha t^{\alpha-1} - 4\pi G\rho_0 t^{\alpha} = 0 \tag{17}$$

$$\alpha(\alpha-1) + \frac{4}{3}\alpha - \frac{2}{3} = 0, \tag{18}$$

where in the second line we plugged in $4\pi G\rho_0 = 3H^2/2$ and canceled out common factors of t. Solving the simple quadratic yields that the growing perturbation solution is $\alpha = 2/3$, so $\delta \sim t^{2/3} \sim a$ in the matter-dominated era. Carrying out similar algebraic manipulation for the radiation-dominated era, where $\rho \sim a^{-4}$, and dark energy-dominated era, where $\rho \sim$ constant, yields $\delta \sim \ln a$ during the radiation epoch and $\delta \sim$ constant during the dark energy epoch. Hence, the perturbations only grow significantly in the matter dominated era, as illustrated by figure 2.

To quantitatively characterize the distribution of large-scale structure in our universe in a way that can be observed, we use the matter power spectrum $P(k)$. The matter power spectrum is defined as the Fourier transform of the two-point matter correlation function $\xi(\vec{r})$:

$$P(k) = \int \xi(\vec{r})e^{-i\vec{k}\cdot\vec{r}}\, d^3\vec{r}. \tag{19}$$

Qualitatively, the correlation function can be thought of as a measure of the matter overdensity relative to the mean density (i.e. the "lumpiness") of a specific patch of the sky. Mathematically, it gives the probability that, given a random galaxy on the sky, another galaxy will be found within a certain length scale of that galaxy, averaged over all initial choices of a first galaxy. This probability is then normalized against the probability that they would be found if the galaxies were simply distributed across the sky in a uniformly random manner. In equation form, this means $\xi(\vec{r})$ is proportional to δ^2, averaged over all positions. Hence we can also express the power spectrum as:

$$P(k) = \langle|\delta_{\vec{k}}|^2\rangle, \tag{20}$$

where $P(k)$ is an average over all orientations of the wave vector $\vec{k}$ and $\delta_{\vec{k}}$ is the Fourier mode corresponding to the real-space perturbation δ [7].

A graph of the matter power spectrum as a function of wave number k is shown in Figure 3. Low k values correspond to large wavelengths and larger scale structures through the relation $k \sim 2\pi/\lambda$. Hence high k values correspond to small scale structures. The spectrum goes like $\sim k$ at low k and goes like $\sim k^{-3}$ at high k. Non-linear clustering, where we no longer assume that the perturbation $\delta \ll 1$, begins to occur at $k \sim 0.1$ h/Mpc, corresponding to a length scale of $\sim$10 Mpc.

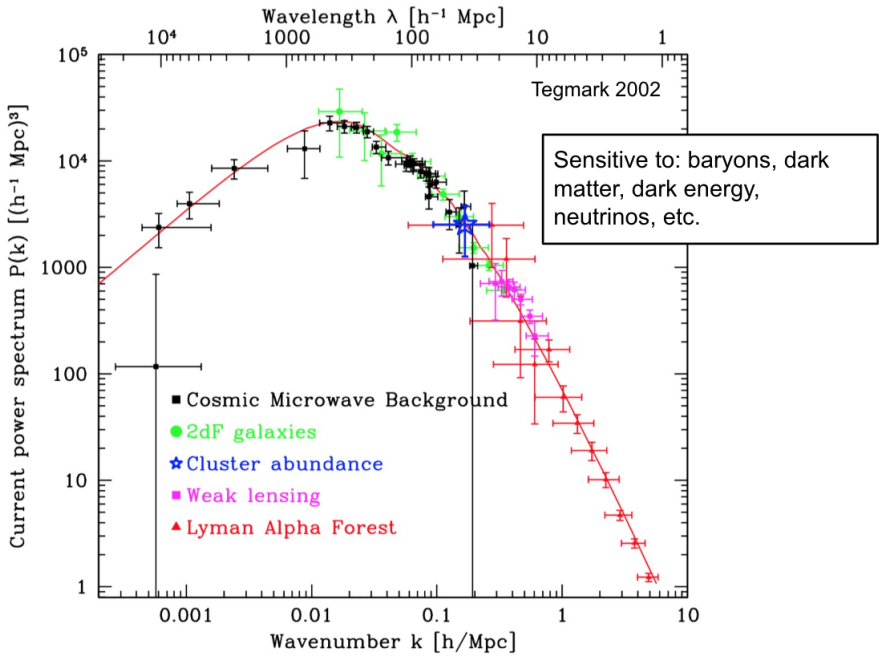

Figure 3: Matter power spectrum at the present-day, $z = 0$. Larger length scales correspond to smaller k, smaller length scales correspond to larger k. Figure taken from [11].

The shape of the matter power spectrum, especially at large k in the non-linear regime, is sensitive to the nature of the matter. For example, because baryonic matter was coupled to radiation (photons) until recombination, it could not form structures before that time. If the ratio of baryonic matter to dark matter were different, then the redshift at which recombination occurs would be different and hence would change when baryonic structures could start growing. This is reflected in the shape of the spectrum. We will focus in particular on how the mass of neutrinos affects the spectrum, thus showing that the matter power spectrum acts a key observable constraint on the neutrino mass.

2.2 Neutrino Effects on Structure Formation

There are two mechanisms through which neutrinos affect the growth of large-scale structure. The first mechanism is that below a certain length scale, neutrinos freely stream over large distances without falling into gravitational potential wells. This cutoff length scale is called the neutrino free-streaming length. Neutrinos freely stream below this length scale because their velocity is higher than the escape velocity of a gravitational potential well of this length scale. Above the scale of the neutrino free streaming length, the neutrinos do cluster just like CDM particles. Since the neutrinos don't cluster, they don't contribute to the growth of structure.

The lack of neutrino clustering at small scales manifests itself in the matter power spectrum at low redshifts ($z \sim 1$) as follows. The total perturbation δ can be split into a perturbation due to neutrinos and a perturbation due to CDM:

$$\delta = \delta_\nu + \delta_{\text{CDM}} = f_\nu \delta + (1 - f_\nu)\delta, \tag{21}$$

where $f_\nu = \Omega_\nu / \Omega_m$ is the ratio of the matter density made up of neutrinos. At small scales, ignoring neutrino clustering gives $P(k) \sim \delta^2 = (1 - f_\nu)^2 \delta^2$. Assuming f_ν is very small, we have that the power spectrum is reduced by a factor of $(1 - 2f_\nu)$.

The second mechanism is that because neutrinos don't cluster at small scales, they cannot contribute to the growth of the perturbations at low redshift. Hence, instead of $\delta \sim a$ in the matter-dominated era, the growth of δ is slightly suppressed to a factor of $\delta \sim a^{1-\frac{3}{5}f_\nu}$. Plugging this behavior into the perturbation equation of motion allows us to derive that the power spectrum will additionally be suppressed by a factor of $(1 - 6f_\nu)$. Hence, the overall effect of neutrinos is to suppress the matter power spectrum by a factor of $(1-2f_\nu)(1-6f_\nu) \approx (1-8f_\nu)$, where we only keep terms to first order. See [12] for a full derivation of this suppression factor. Note that the suppression factor only manifests itself in the power spectrum at small scales; at large scales (small k), the neutrino perturbations are indistinguishable from the CDM perturbations because they can also cluster, meaning measurements of $P(k)$ at large k cannot act as a constraint on the total neutrino mass.

2.3 Dark Matter Halo Formation

In this paper we focus particularly on the effects of massive neutrinos on the growth of dark matter halos. Dark matter halos are the most basic unit of structure in our universe. They are comprised of dark matter and envelop the luminous components of galaxies and galaxy clusters, extending well beyond the visible edge of such objects. Halos form from regions of overdensity (usually approximately 200 times more dense

than the mean density [13]) which become gravitationally bound to each other and decouple from the Hubble expansion, evolving in a non-linear fashion.

The dark matter in halos has never been directly observed. However, their existence is strongly suggested through evidence such as galaxy rotation curves and theoretical predictions of structure growth. Measurements of galaxy rotation curves show that the rotation speed of a galaxy does not drop off past the luminous edge of a galaxy, but rather remains almost constant. This suggests that there must exist additional "dark" matter beyond the visible edge of the galaxy which only interacts gravitationally with regular matter, but not through any other means. Furthermore, if all matter were made of baryons, matter could only start clumping together to make structure at the era of recombination, $z = 1100$, because until then it was coupled with photon radiation. This does not give enough time for the structures we observe today to have formed. Therefore, there must be a dark matter component which does not interact with radiation and is free to gravitationally cluster starting at matter-radiation equality at $z = 3500$.

The distribution of dark matter halos is described using the halo mass function, which is given in [14] as:

$$\frac{dn}{dM} = f(\sigma)\frac{\bar{\rho}_m}{M}\frac{d\ln\sigma^{-1}}{dM}. \tag{22}$$

Here, dn/dM denotes the number of halos per unit volume in the range of mass M to $M + \delta M$, and $\bar{\rho}_m$ denotes the average matter-energy density. The function $f(\sigma)$ is given by:

$$f(\sigma) = A\left[\left(\frac{\sigma}{b}\right)^{-a} + 1\right]e^{-c/\sigma^2}, \tag{23}$$

where σ is given by the formula:

$$\sigma = \int P(k)\hat{W}(kR)k^2 dk. \tag{24}$$

$P(k)$ is the usual matter power spectrum and $\hat{W}$ is the Fourier transform of the real space top-hat window function W of radius R. Qualitatively, W is just used to delimit the volume of a spherical region in space that could potentially contain a halo. A, a, b, and c are parameters whose values are determined by fitting to N-body simulations. Equation 22 was derived in its original form in the Press and Schecter (1974) paper, who used the fact that halos could be associated with peaks in the Gaussian random density field of dark matter in the early universe to obtain a similar functional form [15].

For a given halo mass and fixed mean matter density, the behavior of dn/dM is essentially governed by the $f(\sigma)$ term, since the logarithmic derivative term is expected to be small. From equation 23, we see that the exponential term dominates $f(\sigma)$, and that as σ^2 decreases, the exponential term suppresses $f(\sigma)$ more and more. Since neutrinos suppress $P(k)$ at large k where the length scales are similar to the size of dark

matter halos, we expect that massive neutrinos will cause σ^2 to be suppressed. Thus $f(\sigma)$ and dn/dM will both be lowered. According to Figure 4, the strongest suppression of the matter power spectrum at low redshifts occurs at $k \sim 1$, which corresponds to length scales at the galaxy cluster level. Since galaxy clusters have masses on the order of 10^{14} to $10^{15} M_{\odot}$, we expect halos with this mass range to be suppressed the most by neutrinos.

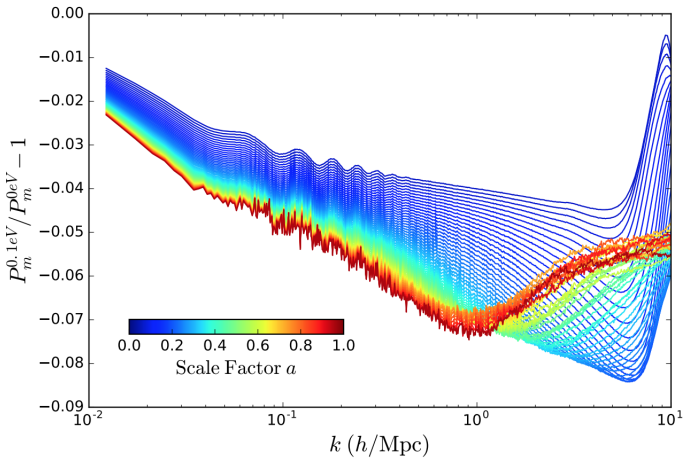

Figure 4: Ratio of matter power spectrum with 0.1 eV neutrino mass to matter power spectrum with massless neutrino, normalized so that a value of less than 0 indicates suppression. At low redshifts, the suppression is strongest around $k \sim 1$, with suppression decreasing at smaller scales. Figure from [16].

3 Cosmological Simulations of Massive Neutrinos

Improving the effectiveness and accuracy of future cosmological surveys requires the ability to computationally model large-scale structure growth and the effects of massive neutrinos on the matter density field extremely accurately. In particular, highly accurate cosmological simulations of massive neutrinos allow cosmologists to confidently compare simulation results obtained from specific initial condition setups with observations of the matter power spectrum in order to constrain the neutrino mass.

However, there currently exists a trade-off between accuracy and computational complexity when simulating neutrino effects on structure growth. In principle, structure growth can be exactly modeled by using traditional N-body cosmological simulations, where cold dark matter and neutrinos are treated as particles. These N-body simulations essentially work by solving for the evolution of discrete phase-space particles with a large thermal velocity imposed on the initial conditions. The particles are evolved over time only under the effects of gravity. The simulation is easy to implement, but it is computationally expensive, with an order of growth of $\mathcal{O}(N \log N)$ and requiring lots of

parallel computing power because of the memory required to store all of the CDM and neutrino particles. Further, because of the randomness introduced into the simulation through the initial conditions, the neutrino power spectrum will contain a scale-invariant white-noise term called shot noise, which is proportional to $1/N$. Because neutrinos do not cluster significantly at scales below their free-streaming length, their power spectrum at the smallest scales (largest k) is dominated by shot noise. This shot noise has an effect on the gravitational potential being used to evolve the particles, which leads to numerical errors that can potentially cause the growth of spurious structures. Of course, one could reduce the shot noise by greatly increasing the number of particles in the simulation, but that would just add to the computational cost of the simulation [17]. Therefore, alternate techniques have been developed to bypass or eliminate the problems of N-body simulation of massive neutrinos, while retaining accuracy. One such alternate technique is to use perturbation theory to model massive neutrino effects, whereby neutrinos are treated as perturbations in Fourier space instead of as particles. The density field of the neutrinos is then calculated using the linear theory. A version of this technique is discussed below.

3.1 Linear Response Approximation

The linear response approximation (LRA) is a computational simulation technique based on perturbation theory that was first implemented by Yacine Ali-Haïmoud and Simeon Bird in [18], hereafter referred to as AHB13. This approach is more accurate than pure linear perturbation theory because it also accounts for the full non-linear CDM potential that becomes important during non-linear structure growth [17]. Therefore the technique could be more accurately described as being semi-linear. It models the non-linear growth of CDM clusters and gravitational potentials very well, while using a linear transfer function to obtain neutrino overdensities from the gravitational potential [18]. All of this is done with no extra computational cost. In fact, the LRA is better than traditional particle methods computationally because it requires less memory to store all the information about the particles, since the neutrinos are not treated as particles. AHB13 found that the LRA allows for an accurate modeling of the matter power spectrum up to the present time, and for redshift of $z \geq 1$, an accurate modeling of the neutrino power spectrum as well; however, LRA fails to resolve neutrino clustering in the most massive galaxy clusters.

Quantitatively, AHB13 showed that the LRA very effectively modeled the interactions between neutrinos and CDM, computing the matter power spectrum to 99.8% accuracy for a total neutrino mass of $M < 0.6$ eV. However, for $M > 0.3$ eV and redshift $z < 0.5$, LRA underestimates neutrino clustering relative to the N-body simulation. This is because LRA is at its core a perturbative approximation, involving an expansion to first order in ϕ/v^2, where ϕ denotes the gravitational potential and v denotes the

velocity of the neutrino. Since this quantity is no longer small when v becomes small, the LRA fails in the limit of low initial velocities. Neutrinos are fermions that obey Fermi-Dirac statistics (equation 1), so there will always be a certain fraction of neutrinos that have small enough velocities to invalidate the LRA, or in other words, cluster non-linearly from the beginning. These slow initial velocity neutrinos can dominate clustering at small scales, even though they make up only a small fraction of the total neutrino distribution. Therefore, while LRA serves as a good tool to use to simulate the evolution of cold dark matter in our universe, it still has some flaws when simulating the evolution of neutrinos and how they interact with the dark matter.

3.2 The Hybrid Method

The hybrid method, presented in [17], serves as an improvement over the LRA. It follows the LRA model of using linear response approximation to simulate the neutrinos and N-body code to simulate the CDM particles. However, the difference between the hybrid method and the original LRA is that in the hybrid method, the massive neutrinos are split into fast and slow components. The fast component is simulated from $z = 99$ to $z = 0$ fully with the LRA, whereas the slow component is only simulated with LRA from $z = 99$ to $z = 1$. From $z = 1$ to $z = 0$, the slow component is simulated using the N-body particle method by treating the neutrinos as particles, thereby capturing the non-linear clustering behavior of the neutrinos. Importantly, the slow neutrinos are still initialized at $z = 99$ so that they will have accurate positions and velocities at $z = 1$ when they are switched on to particle mode. However, they are not used to calculate the gravitational potential until $z = 1$; instead, they evolve only according to the potential defined by the CDM and linear response neutrinos, so they effectively behave as tracer particles.

Note that the hybrid method could also have been implemented by only introducing the slow neutrinos at $z = 1$ and using the LRA to analytically calculate what their distribution and velocity would be at that moment. However, this version of the implementation is more difficult than inserting the slow neutrinos as tracers at $z = 99$ because computing the velocities with LRA at $z = 1$ is decidedly more difficult than taking the initial velocities at $z = 99$ from the usual Fermi-Dirac distribution. Therefore in [17] they used the implementation of initialization at $z = 99$.

The advantages of the hybrid model over both the LRA and the usual N-body particle model are as follows. Because the particle behavior of the slow neutrinos are only switched on at $z = 1$, as opposed to having the slow neutrinos behave as particles for all redshifts, the worst effects of shot noise are mitigated. Further, introducing particle behavior for slow neutrinos at $z = 1$ allows us to fully capture the non-linear clustering of slow velocity neutrinos better than the LRA. Preserving the use of LRA

for all redshifts $z > 1$ allows us to accurately capture the linear behavior of the neutrino perturbations at earlier times, as desired. In the first part of our results, we analyze the results of simulation runs using the hybrid method and compare them to particle-based and LRA-based simulations.

3.3 Simulation Details

All simulations were run under the following conditions. The size of each simulation was a cube with side length 300 Mpc/h, co-moving at the time of $z = 99$. h is simply a non-dimensional parameter such that $100h$ equals the current value of the Hubble parameter. Each box was initially filled with 512^3 CDM particles distributed randomly and with nearly zero initial velocity. For the simulations that treated neutrinos as particles, each box was also filled with 512^3 neutrino particles. Neutrino initial velocities were assigned based on the Fermi-Dirac probability distribution in Equation 1. CDM and neutrinos were then evolved under the conditions explained in the previous two subsections from $z = 99$ to $z = 0$, with their motions only being governed by their self-gravitational attraction, thus forming dark matter halos.

To determine the center-of-mass and position of the halos that developed in our simulation runs, we used the Friend-of-Friend (FOF) halo finder algorithm. A detailed explanation of how the algorithm works is given in [19], but we will give a brief qualitative description here. FOF works by defining a linking length. Particles that are closer to each other than the linking length are considered "linked", with the links developing as one group initializes development and looks for more particles to join. When different hierarchies of linking lengths are defined, distinct overdensity regions can be established, through which halos can be distinguished. The hierarchy of linking lengths also allows for subhalos to be defined inside of halos. By the constraints of FOF, subhalos must lie completely inside their parent halo. Furthermore, FOF halos cannot intersect because each particle can only uniquely belong to one halo at most. Therefore, if two halos are separated by less than the linking length, they will be joined to become one large halo. In all cases, halos are identified through the usual definition of being a region that is at least 200 times more dense than the average density.

4 Results and Discussion

Data from the simulations was analyzed at two different redshifts: $z = 0.2$, which approximates the structure of our universe today because it is in the dark-energy dominated era already, and $z = 1$, which represents the state of the universe approximately 3 billion years before matter-dark energy equality. In section 4.1, we present the results of validity tests on the Hybrid method. In section 4.2, we present the results of a halo

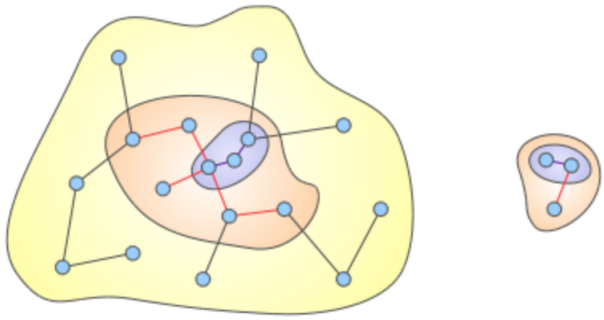

Figure 5: A schematic depiction of how the Friend-of-Friend (FOF) halo finder works to find dark matter halos. If different linking lengths are defined, then a hierarchy of halo substructures at different overdensity levels can be established, as shown by the purple, orange, and yellow layers. If the two distinct halos get close enough, FOF will join them into one large halo. Image taken from https://www.cosmosim.org/cms/simulations/halo-finders/fof.

mass function analysis using the simulation data. In all plots and analyses presented below, $\log M_\odot$ denotes the logarithm base 10 of the mass, where $M_\odot$ is 1 solar mass.

4.1 Validity of the Hybrid Method

First we tested the validity of the Hybrid Method in simulating dark matter halo growth in the universe. To that end, we took the results of the particle-based simulation to be the "truth" simulation, since that is what is traditionally used in cosmological simulations. We then compared the results of the Hybrid method simulation runs with the results of the LRA method simulation runs, relative to the truth results.

Using the same initial conditions, we ran particle, LRA, and hybrid-based simulations of structure growth, assuming a 0.4 eV neutrino mass cosmology. Once the mass and position of the center-of-mass of each halo was found with FOF, we used a Nearest Neighbor algorithm implemented using the kd-tree structure in `Python` to match the distances between the centers-of-masses of seemingly identical halos in each simulation. The kd-tree for each simulation was built by inserting the 3-D position of the center-of-mass of each halo into a binary search tree structure, with insertion based upon the position of each halo relative to the current halos that make up the nodes of the tree. Once the tree was constructed, nearest neighbor searches could be performed by traversing down the tree, pruning particular branches of the tree if it is known that the nearest neighbor halo could not be in a particular region of the simulation space. The results of this cross-matching are presented in Figure 6.

From Figure 6 we see that the overwhelming majority of matching halos were found within 200 kpc of each other, with an exponential decrease in number of matched

halos found at distances of more than 200 kpc apart. This indicates that the LRA and hybrid methods each individually performed well in creating halos located very close to where they would have been located in the particle method. Further, the high degree of matching between the two methods at both redshifts indicates the hybrid method was as successful as the LRA method, up to separation distances of 10 Mpc.

To doubly confirm that the halos cross-matched by distance were truly as identical as possible, we compared the masses of the cross-matched halos. If the ratio of the masses of the cross-matched halos was 1, that indicated a perfect match in terms of mass. Because of the exponential decrease in number of cross-matched halos as separation distance increased, we split the cross-matched halos into 4 categories based on their separation distance: 0-50 kpc, 50-100 kpc, 100-200 kpc, and >200 kpc. The results of this analysis are presented in Figures 7 and 8.

We see that in nearly every single case, the distribution of the mass ratio was centered around 1, with the peak becoming sharper for higher halo masses. Further, there was good agreement between the LRA method ratio distribution and hybrid method ratio distribution across all separation distances and mass ranges. This confirms that the hybrid method produces accurate results in generating dark matter halos.

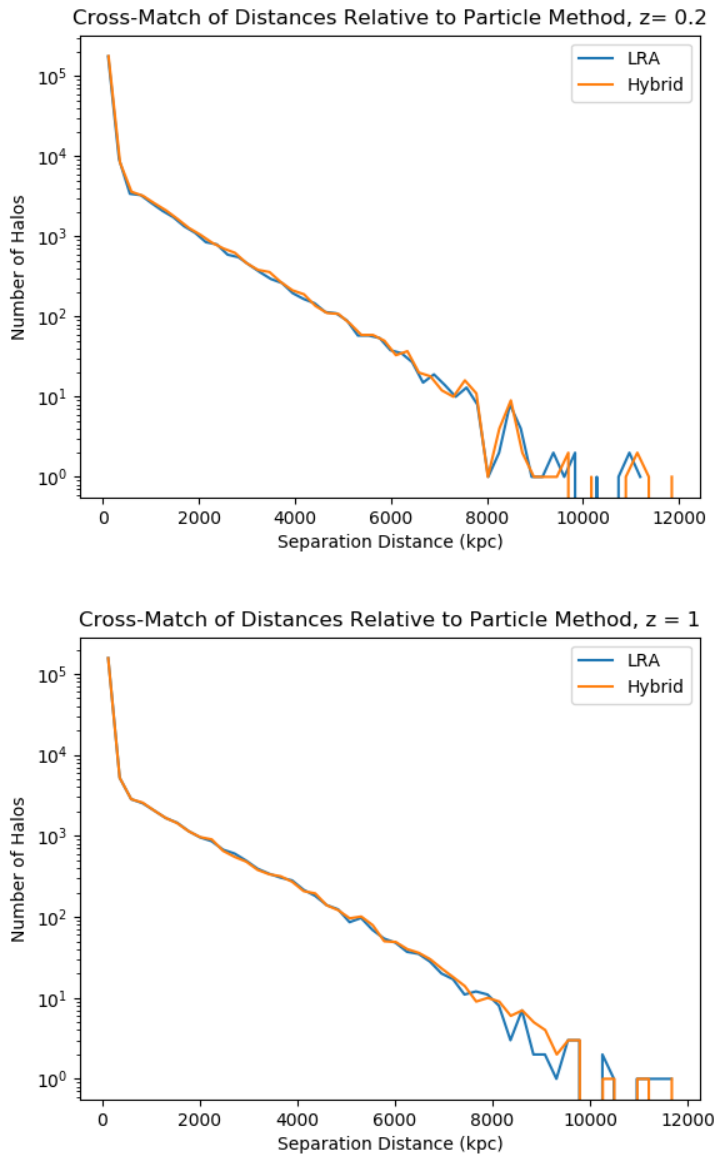

Figure 6: Cross-matched distance between halos of the particle ("truth") simulation and the LRA and Hybrid simulations, respectively. Data comes from using a 0.4 eV neutrino mass, and was analyzed at $z = 0.2$ and $z = 1$.

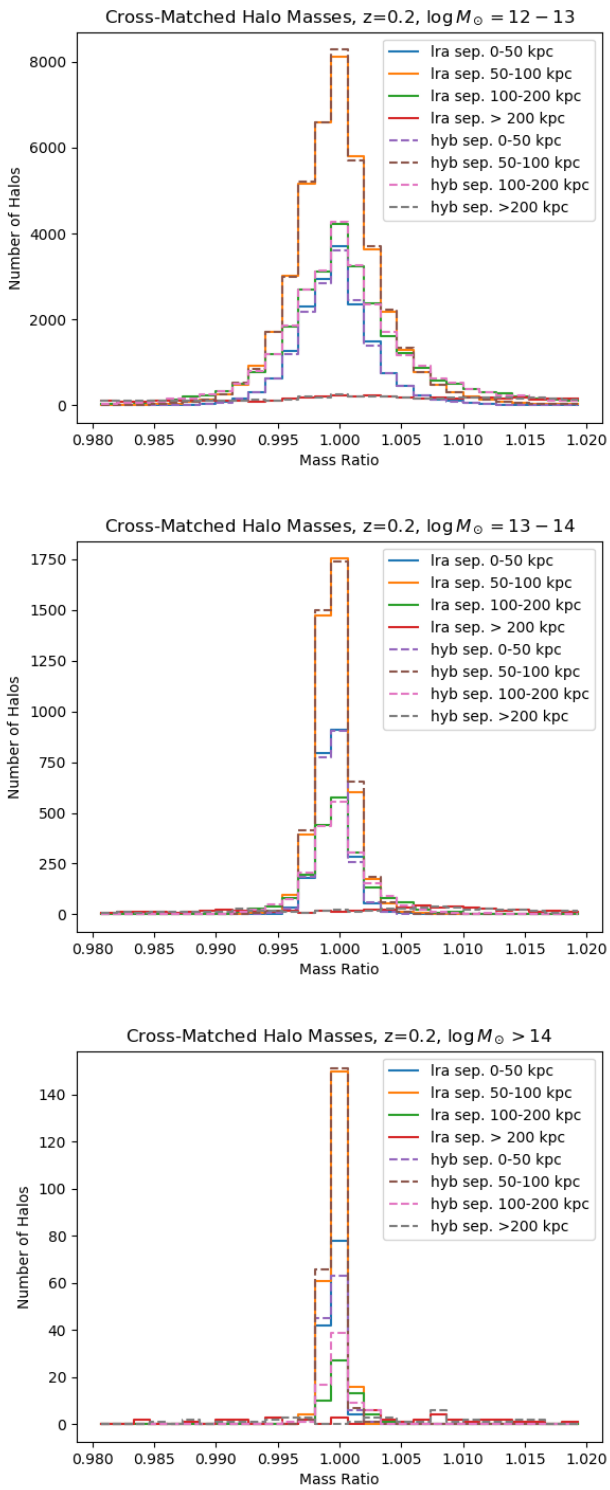

Figure 7: Cross-matched halo masses between halos of the particle ("truth") simulation and the LRA and Hybrid simulations, respectively, at redshift $z = 0.2$. Top left plot contains halos with masses between $10^{12} M_\odot$ and $10^{13} M_\odot$. Top right plot contains halos with masses between 10^{13} and $10^{14} M_\odot$. Bottom plot contains halos with masses in excess of $10^{14} M_\odot$.

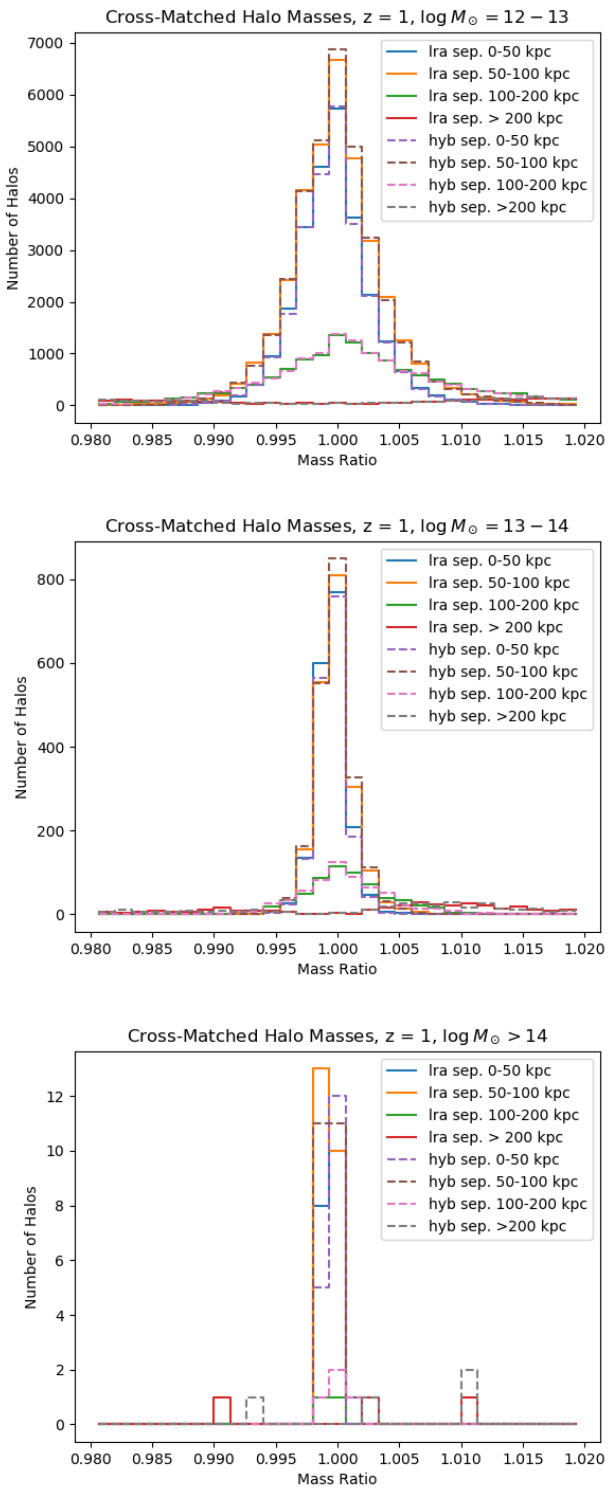

Figure 8: Cross-matched halo masses between halos of the particle ("truth") simulation and the LRA and Hybrid simulations, respectively, at redshift $z = 1$. Top left plot contains halos with masses between $10^{12} M_\odot$ and $10^{13} M_\odot$. Top right plot contains halos with masses between 10^{13} and $10^{14} M_\odot$. Bottom plot contains halos with masses in excess of $10^{14} M_\odot$.

4.2 Halo Mass Function Analysis

First we present the results of halo mass function analysis at redshift $z = 0.2$. We calculated the halo mass function of the halos generated by the hybrid simulation by constructing a simple histogram of counts of number of halos found at particular masses. We then divided out this count by the volume of the simulation box, 300 $\mathrm{Mpc}^3/\mathrm{h}^3$ and the width of each logarithmic mass bin to obtain the number density of halos per logarithmic mass. The results are shown in Figure 9. In comparison, we present the calculated halo mass function for the simulation at $z = 1$ in Figure 10, where the calculation of the mass function was the same as described above.

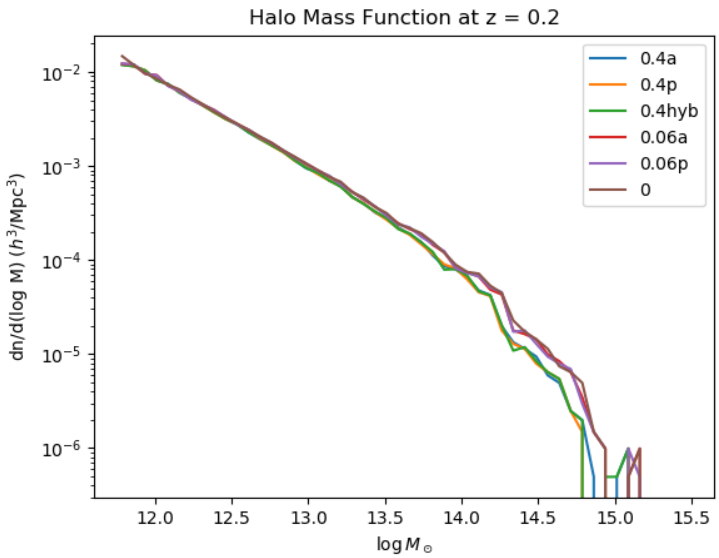

Figure 9: The halo mass function of the dark matter halos generated by all 3 types of simulations at redshift z = 0.2. Neutrino masses of 0 eV, 0.06 eV, and 0.4 eV were used. a represents LRA method, p represents N-body particle method, hyb represents hybrid method.

As expected, at higher redshift, there are less massive halos compared to lower redshift. For example, the number density of halos with mass $> 10^{14.5} M_\odot \approx 3 \times 10^{14} M_\odot$ is nearly zero for all neutrino mass cosmologies and simulation methods at $z = 1$, compared to a number density on the order of $\sim 10^{-5}$ for $z = 0.2$. We also notice that at both $z = 0.2$ and $z = 1$, nonzero neutrino mass effects only increasingly made their presence known at higher and higher halo masses, although the halo mass at which the effects start to be apparent seems to be slightly different between the two redshifts.

To better distinguish the suppression of large halo masses by nonzero neutrino mass, we also present plots of the ratio of halo number density per logarithmic mass between the different neutrino mass cosmologies, for both redshift $z = 0.2$ and $z = 1$. We define this ratio as the suppression ratio, and take it to be the ratio between a 0.4

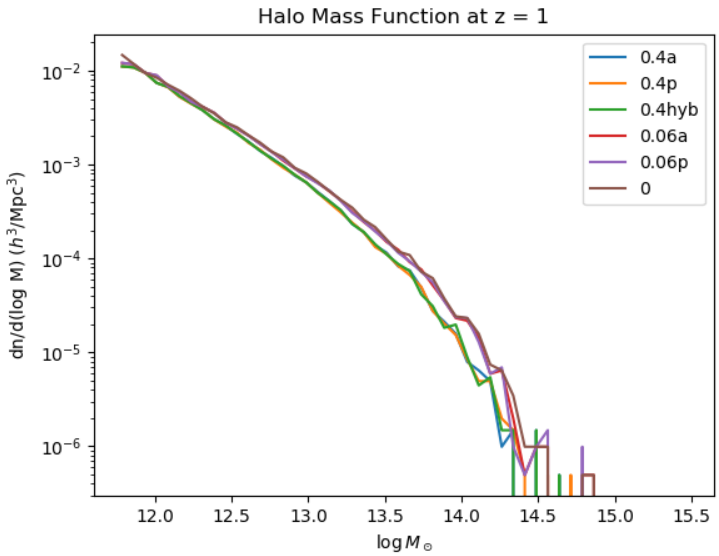

Figure 10: The halo mass function of the dark matter halos generated by all 3 types of simulations at redshift z = 1. Neutrino masses of 0 eV, 0.06 eV, and 0.4 eV were used. a represents LRA method, p represents N-body particle method, hyb represents hybrid method.

eV neutrino cosmology and a massless neutrino cosmology, minus 1. The plots are given in Figures 11 and 12. The dotted line at 0 indicates that if neutrino mass were to have no effect on halo mass, the curve should follow that line.

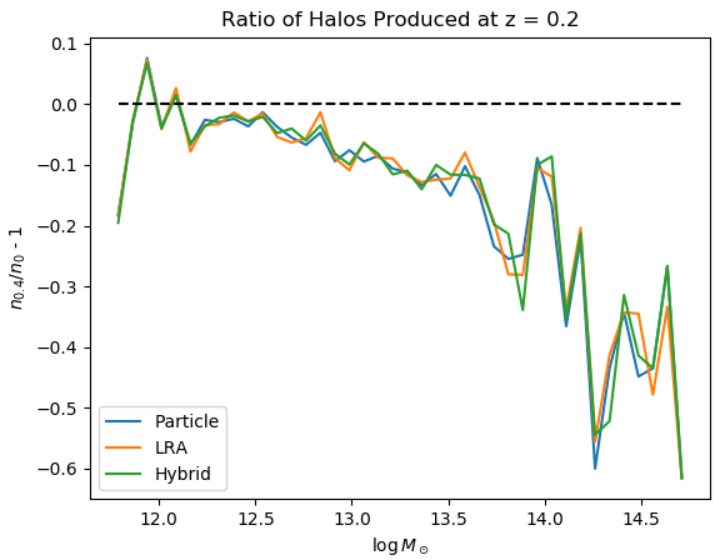

Figure 11: Suppression ratio of halos produced among the different neutrino mass cosmologies (0.4 eV versus 0 eV), at redshift $z = 0.2$. Dotted line at 0 indicates a perfect 1-to-1 ratio, i.e. that neutrino mass would have no effect on halo mass function.

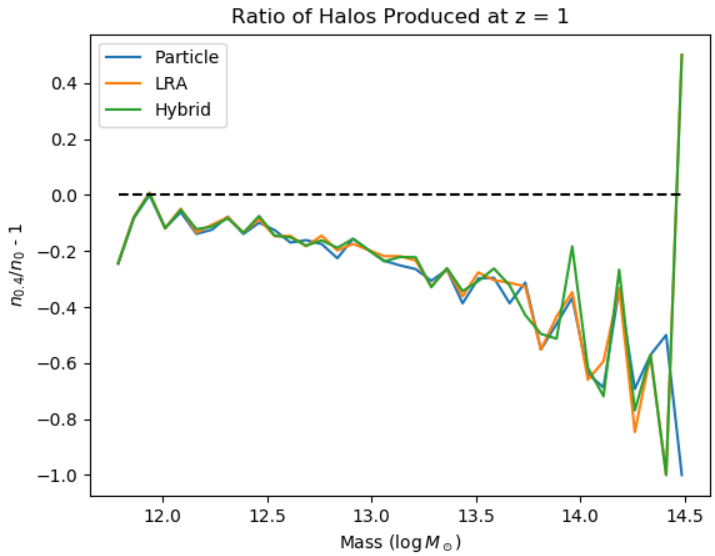

Figure 12: Suppression ratio of halos produced among the different neutrino mass cosmologies (0.4 eV versus 0 eV), at redshift 1. Dotted line at 0 indicates a perfect 1-to-1 ratio, i.e. neutrino mass has no effect on halo mass.

From these plots, we confirm our conclusion that higher mass halos are increasingly suppressed by a nonzero neutrino mass component, since all three methods of simulation at both redshifts show nearly identical results. This result agrees with the theory discussed in section 3.3. The suppression of high mass halos also agrees with the results in [20], which also used different mass neutrinos to investigate their effects on the halo mass function. However, we cannot directly compare the results of Figure 11 to Figure 12 precisely because they occur at different redshifts. That is, halos with masses of $10^{13} M_\odot$ at $z = 1$ might have evolved to halos with masses of $10^{14} M_\odot$ at $z = 0.2$. If we account for the shift in time and its effect on the evolution of halo growth, we see that the results make sense, because suppression beginning at $10^{12} M_\odot$ during $z = 1$ would correspond to suppression beginning at some higher mass, like $10^{13} M_\odot$, at later redshift of $z = 0.2$.

5 Conclusions and Future Work

In this junior paper, we explored the effects of neutrino mass on the suppression of large scale structure growth, specifically focusing on the growth of dark matter halos. Using three different neutrino cosmologies of $M = 0$ eV, $M = 0.06$ eV, $M = 0.4$ eV and three different simulation methods (particle-based, LRA, and hybrid), we first verified that the hybrid method worked extremely well in producing the same halos as those produced by the particle-based and LRA simulation methods, given the same initial

conditions. The verification involved a two-step process of first cross-checking matching halos by their separation distance between center-of-mass, followed by a cross-check of mass ratio between candidate matching halos. Results show that the most common separation distance between matching halos was between $50 - 100$ kpc and the most common mass ratio was exactly 1. To put in context, the typical radius of an average halo is on the order of 100 kpc, so the matched halos were found virtually on top of each other.

We then used all three simulation methods to simulate dark matter halo growth in the presence of massive neutrinos from $z = 99$ to $z = 0$ and analyzed the data to numerically obtain the halo mass function at redshifts $z = 1$ and $z = 0.2$. We found that less massive halos were less suppressed, with suppression only becoming evident at masses of $\gtrsim 10^{13} M_\odot$ for $z = 1$ and $\gtrsim 6 \times 10^{13} M_\odot$ for $z = 0.2$. The suppression ratio reached -0.5 (meaning only half as many halos of that mass had formed as would have formed if neutrinos were massless) at a mass of $10^{14} M_\odot$ for $z = 1$ and $2 \times 10^{14} M_\odot$ for $z = 0.2$. This indicates that suppression has noticeable effects starting at the scale of massive galaxies, with the suppression only increasing as the mass scale increased to galaxy cluster level.

The observed results agreed with theoretical predictions based on the nature of the halo mass function defined in equation 22 and the effect of neutrino mass on the matter power spectrum. We thus tentatively conclude that massive neutrinos suppress the growth of dark matter halos at the scale of large galaxies and above, meaning the quantity of large galaxies and galaxy clusters is reduced at the present-day due to the effects of neutrinos.

Based on these results, there are many paths we can take in the future to further explore the work done in this paper. First and foremost, we can improve the halo finder algorithm used to find all halos in our simulation. Whereas FOF gives the mass and center-of-mass position of each halo, we would like to use the Rockstar halo finder to further classify the found halos based on their shape (whether spherical or elongated) and their spin. In addition, we would like to use these results to explore the degeneracy in σ_8 with respect to neutrino mass and the matter power spectrum. Here σ_8 is defined as the rms of the matter fluctuations in the power spectrum at the scale of 8 Mpc h^{-1}. In other words, how would altering σ_8 in the initial conditions affect halo mass growth and would it do so in similar ways as neutrino mass? Finally we can check how the suppression of the halo mass function translates into a suppression of the matter power spectrum itself at the smallest scales and compare the results with observations, thus giving a more robust constraint on the absolute neutrino mass.

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